Table 5.2.2.2.1-10: Codebook for 6-layer CSI reporting using antenna ports 3000 to 2999+$P_{\text{CSI-RS}}$

***codebookMode* = 1-2**				
	$i_{1,1}$	$i_{1,2}$	i_2	
$N_2 > 1$	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1	$W^{(6)}_{i_{1,1},i_{1,1}+O_1,i_{1,1}+O_1,i_{1,2},i_{1,2}+O_2,i_2}$
$N_1 > 2, N_2 = 1$	$0,\ldots,N_1O_1-1$	0	0,1	$W^{(6)}_{i_{1,1},i_{1,1}+O_1,i_{1,1}+2O_1,0,0,0,i_2}$
where $W^{(6)}_{l,l',l'',m,m'.m'',n} = \frac{1}{\sqrt{6P_{\text{CSI-RS}}}}\begin{bmatrix} v_{l,m} & v_{l,m} & v_{l',m'} & v_{l',m'} & v_{l'',m''} & v_{l'',m''} \\ \varphi_n v_{l,m} & -\varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l',m'} & v_{l'',m''} & -v_{l'',m''} \end{bmatrix}$.				

Table 5.2.2.2.1-11: Codebook for 7-layer CSI reporting using antenna ports 3000 to 2999+$P_{\text{CSI-RS}}$

***codebookMode* = 1-2**				
	$i_{1,1}$	$i_{1,2}$	i_2	
$N_1 = 4, N_2 = 1$	$0,\ldots,\frac{N_1O_1}{2}-1$	0	0,1	$W^{(7)}_{i_{1,1},i_{1,1}+O_1,i_{1,1}+2O_1,i_{1,1}+3O_1,0,0,0,0,i_2}$
$N_1 > 4, N_2 = 1$	$0,\ldots,N_1O_1-1$	0	0,1	$W^{(7)}_{i_{1,1},i_{1,1}+O_1,i_{1,1}+2O_1,i_{1,1}+3O_1,0,0,0,0,i_2}$
$N_1 = 2, N_2 = 2$	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1	$W^{(7)}_{i_{1,1},i_{1,1}+O_1,i_{1,1},i_{1,1}+O_1,i_{1,2},i_{1,2},i_{1,2}+O_2,i_{1,2}+O_2,i_2}$
$N_1 > 2, N_2 = 2$	$0,\ldots,N_1O_1-1$	$0,\ldots,\frac{N_2O_2}{2}-1$	0,1	$W^{(7)}_{i_{1,1},i_{1,1}+O_1,i_{1,1},i_{1,1}+O_1,i_{1,2},i_{1,2},i_{1,2}+O_2,i_{1,2}+O_2,i_2}$
$N_1 > 2, N_2 > 2$	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1	$W^{(7)}_{i_{1,1},i_{1,1}+O_1,i_{1,1},i_{1,1}+O_1,i_{1,2},i_{1,2},i_{1,2}+O_2,i_{1,2}+O_2,i_2}$
where $W^{(7)}_{l,l',l'',l''',m,m'.m'',m''',n} = \frac{1}{\sqrt{7P_{\text{CSI-RS}}}}\begin{bmatrix} v_{l,m} & v_{l,m} & v_{l',m'} & v_{l'',m''} & v_{l'',m''} & v_{l''',m'''} & v_{l''',m'''} \\ \varphi_n v_{l,m} & -\varphi_n v_{l,m} & \varphi_n v_{l',m'} & v_{l'',m''} & -v_{l'',m''} & v_{l''',m'''} & -v_{l''',m'''} \end{bmatrix}$.				

Table 5.2.2.2.1-12: Codebook for 8-layer CSI reporting using antenna ports 3000 to 2999+$P_{\text{CSI-RS}}$

***codebookMode* = 1-2**				
	$i_{1,1}$	$i_{1,2}$	i_2	
$N_1 = 4, N_2 = 1$	$0,\ldots,\frac{N_1O_1}{2}-1$	0	0,1	$W^{(8)}_{i_{1,1},i_{1,1}+O_1,i_{1,1}+2O_1,i_{1,1}+3O_1,0,0,0,0,i_2}$
$N_1 > 4, N_2 = 1$	$0,\ldots,N_1O_1-1$	0	0,1	$W^{(8)}_{i_{1,1},i_{1,1}+O_1,i_{1,1}+2O_1,i_{1,1}+3O_1,0,0,0,0,i_2}$
$N_1 = 2, N_2 = 2$	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1	$W^{(8)}_{i_{1,1},i_{1,1}+O_1,i_{1,1},i_{1,1}+O_1,i_{1,2},i_{1,2},i_{1,2}+O_2,i_{1,2}+O_2,i_2}$
$N_1 > 2, N_2 = 2$	$0,\ldots,N_1O_1-1$	$0,\ldots,\frac{N_2O_2}{2}-1$	0,1	$W^{(8)}_{i_{1,1},i_{1,1}+O_1,i_{1,1},i_{1,1}+O_1,i_{1,2},i_{1,2},i_{1,2}+O_2,i_{1,2}+O_2,i_2}$
$N_1 > 2, N_2 > 2$	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1	$W^{(8)}_{i_{1,1},i_{1,1}+O_1,i_{1,1},i_{1,1}+O_1,i_{1,2},i_{1,2},i_{1,2}+O_2,i_{1,2}+O_2,i_2}$
where $W^{(8)}_{l,l',l'',l''',m,m'.m'',m''',n} = \frac{1}{\sqrt{8P_{\text{CSI-RS}}}}\begin{bmatrix} v_{l,m} & v_{l,m} & v_{l',m'} & v_{l',m'} & v_{l'',m''} & v_{l'',m''} & v_{l''',m'''} & v_{l''',m'''} \\ \varphi_n v_{l,m} & -\varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l',m'} & v_{l'',m''} & -v_{l'',m''} & v_{l''',m'''} & -v_{l''',m'''} \end{bmatrix}$.				

5.2.2.2.1a Refined Type I Single-Panel Codebook

For 48, 64 and 128 antenna ports, obtained by aggregating K CSI-RS resources for channel measurement, and the UE configured with higher layer parameter *codebookType* set to 'typeI-SinglePanel-r19'

- The values of N_1 and N_2 are configured with the higher layer parameter *n1-n2-typeI-r19*. The supported configurations of (N_1, N_2), for a given number of CSI-RS ports, are given in Table 5.2.2.2.1a-1, for which $O_1 = O_2 = 4$. The number of CSI-RS ports, P_{CSI-RS}, is $2N_1N_2$.

Table 5.2.2.2.1a-1: Supported configurations of (N_1, N_2)

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1, N_2)
48	(8,3)
	(6,4)
64	(16,2)
	(8,4)
128	(16,4)
	(8,8)

- The port index, $p' = 0,1,\ldots,P_{CSI-RS}-1$, is described in Clause 7.4.1.5.3 of [4, TS 38.211].

For the UE configured with higher layer parameter *typeI-codebookMode-r19* set to 'modeA', each PMI value corresponds to the codebook indices i_1, i_2, where the composite codebook index i_1, for reported rank υ, is defined by

$$i_1 = \begin{cases} [i_{1,1} \quad i_{1,2}] & \upsilon = 1 \\ [i_{1,1} \quad i_{1,2} \quad i_{1,3}] & \upsilon = 2,3,4 \\ [i_{1,1} \quad i_{1,2} \quad i_{1,3} \quad i_{1,1,2} \quad i_{1,2,2} \quad i_{1,1,3} \quad i_{1,2,3}] & \upsilon = 5,6 \\ [i_{1,1} \quad i_{1,2} \quad i_{1,3} \quad i_{1,1,2} \quad i_{1,2,2} \quad i_{1,1,3} \quad i_{1,2,3} \quad i_{1,1,4} \quad i_{1,2,4}] & \upsilon = 7,8 \end{cases}$$

The codebook for 1-8 layers is given in Table 5.2.2.2.1a-4 with the value range of i_1 and i_2 given in Table 5.2.2.2.1a-2. The mapping from $i_{1,3}$ to k_1 and k_2 for 2-layer, 3-layer and 4-layer reporting, and to (o_1, k_1) and (o_2, k_2) for 5-layer, 6-layer, 7-layer and 8-layer reporting is given in Table 5.2.2.2.1a-3. The quantities φ_n, u_m, t_l and $v_{l,m}$ are given by

$$\varphi_n = e^{j\frac{\pi n}{2}}$$

$$u_m = \begin{bmatrix} 1 & e^{j\frac{2\pi m}{O_2 N_2}} & \ldots & e^{j\frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix}, \quad t_l = \begin{bmatrix} 1 & e^{j\frac{2\pi l}{O_1 N_1}} & \ldots & e^{j\frac{2\pi l(N_1-1)}{O_1 N_1}} \end{bmatrix}$$

$$v_{l,m} = \begin{bmatrix} u_m & e^{j\frac{2\pi l}{O_1 N_1}} u_m & \ldots & e^{j\frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T$$

The reported values of $(i_{1,1}, i_{1,2})$, $(i_{1,1,2}, i_{1,2,2})$, $(i_{1,1,3}, i_{1,2,3})$, $(i_{1,1,4}, i_{1,2,4})$ correspond to vectors $v_{l,m}, v_{l',m'}, v_{l'',m''}, v_{l''',m'''}$ that are orthogonal in at least one dimension such that for any pair of vectors, v_{l_1,m_1} and v_{l_2,m_2}, with $l_1, l_2 \in \{l, l', l'', l'''\}$, $l_1 \neq l_2$, and $m_1, m_2 \in \{m, m', m'', m'''\}$, $m_1 \neq m_2$, u_{m_1} and u_{m_2} are orthogonal and/or t_{l_1} and t_{l_2} are orthogonal.

Table 5.2.2.2.1a-2: Value range of i_1 and i_2 for 1-8 layers and *typeI-codebookMode-r19* = 'modeA'

Layers	*typeI-codebookMode-r19* = 'modeA'					
$\upsilon = 1$	$i_{1,1}$	$i_{1,2}$				i_2
	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$				0,1,2,3
$\upsilon = 2,3,4$	$i_{1,1}$	$i_{1,2}$	$i_{1,3}$			i_2
	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1,2,3			0,1
$\upsilon = 5,6,7,8$	$i_{1,1}$	$i_{1,2}$	$i_{1,3}$	$i_{1,1,2}$ $i_{1,1,3}$ $i_{1,1,4}$	$i_{1,2,2}$ $i_{1,2,3}$ $i_{1,2,4}$	i_2
	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0	$0,\ldots,N_1-1$	$0,\ldots,N_2O_2-1$	0,1
			1	$0,\ldots,N_1O_1-1$	$0,\ldots,N_2-1$	

Table 5.2.2.2.1a-3: Mapping of $i_{1,3}$ to k_1 and k_2 for 2-4 layers, and to (o_1, k_1) and (o_2, k_2) for 5-8 layers and *typeI-codebookMode-r19* = 'modeA'

$i_{1,3}$	$\upsilon = 2$		$\upsilon = 3,4$		$\upsilon = 5,6,7,8$	
	k_1	k_2	k_1	k_2	(o_1, k_1)	(o_2, k_2)
0	0	0	O_1	0	$(O_1, \mathrm{mod}(i_{1,1}, O_1))$	$(1,0)$
1	O_1	0	0	O_2	$(1,0)$	$(O_2, \mathrm{mod}(i_{1,2}, O_2))$
2	0	O_2	O_1	O_2		
3	$2O_1$	0	$2O_1$	0		

Table 5.2.2.2.1a-4: Codebook for 1-8 layers and *typeI-codebookMode-r19* = 'modeA'

Layers	*typeI-codebookMode-r19* = 'modeA'
$\upsilon = 1$	$W^{(1)}_{i_{1,1},i_{1,2},i_2}$
	where $W^{(1)}_{l,m,n} = \frac{1}{\sqrt{P_{CSI-RS}}}\begin{bmatrix} v_{l,m} \\ \varphi_n v_{l,m} \end{bmatrix}$
$\upsilon = 2$	$W^{(2)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_2}$
	where $W^{(2)}_{l,l',m,m',n} = \frac{1}{\sqrt{2P_{CSI-RS}}}\begin{bmatrix} v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1a-3
$\upsilon = 3$	$W^{(3)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_2}$
	where $W^{(3)}_{l,l',m,m',n} = \frac{1}{\sqrt{3P_{CSI-RS}}}\begin{bmatrix} v_{l,m} & v_{l',m'} & v_{l,m} \\ \varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l,m} \end{bmatrix}$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1a-3
$\upsilon = 4$	$W^{(4)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_2}$
	where $W^{(4)}_{l,l',m,m',n} = \frac{1}{\sqrt{4P_{CSI-RS}}}\begin{bmatrix} v_{l,m} & v_{l',m'} & v_{l,m} & v_{l',m'} \\ \varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l,m} & -\varphi_n v_{l',m'} \end{bmatrix}$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1a-3
$\upsilon = 5$	$W^{(5)}_{i_{1,1},o_1 i_{1,1,2}+k_1,o_1 i_{1,1,3}+k_1,i_{1,2},o_2 i_{1,2,2}+k_2,o_2 i_{1,2,3}+k_2,i_2}$
	where $W^{(5)}_{l,l',l'',m,m',m'',n} = \frac{1}{\sqrt{5P_{CSI-RS}}}\begin{bmatrix} v_{l,m} & v_{l,m} & v_{l',m'} & v_{l',m'} & v_{l'',m''} \\ \varphi_n v_{l,m} & -\varphi_n v_{l,m} & v_{l',m'} & -v_{l',m'} & v_{l'',m''} \end{bmatrix}$ and the mapping from $i_{1,3}$ to (o_1, k_1) and (o_2, k_2) is given in Table 5.2.2.2.1a-3
$\upsilon = 6$	$W^{(6)}_{i_{1,1},o_1 i_{1,1,2}+k_1,o_1 i_{1,1,3}+k_1,i_{1,2},o_2 i_{1,2,2}+k_2,o_2 i_{1,2,3}+k_2,i_2}$
	where $W^{(6)}_{l,l',l'',m,m',m'',n} =$ $\frac{1}{\sqrt{6P_{CSI-RS}}}\begin{bmatrix} v_{l,m} & v_{l,m} & v_{l',m'} & v_{l',m'} & v_{l'',m''} & v_{l'',m''} \\ \varphi_n v_{l,m} & -\varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l',m'} & v_{l'',m''} & -v_{l'',m''} \end{bmatrix}$ and the mapping from $i_{1,3}$ to (o_1, k_1) and (o_2, k_2) is given in Table 5.2.2.2.1a-3
$\upsilon = 7$	$W^{(7)}_{i_{1,1},o_1 i_{1,1,2}+k_1,o_1 i_{1,1,3}+k_1,o_1 i_{1,1,4}+k_1,i_{1,2},o_2 i_{1,2,2}+k_2,o_2 i_{1,2,3}+k_2,o_2 i_{1,2,4}+k_2,i_2}$
	where $W^{(7)}_{l,l',l'',l''',m,m',m'',m''',n} =$ $\frac{1}{\sqrt{7P_{CSI-RS}}}\begin{bmatrix} v_{l,m} & v_{l,m} & v_{l',m'} & v_{l'',m''} & v_{l'',m''} & v_{l''',m'''} & v_{l''',m'''} \\ \varphi_n v_{l,m} & -\varphi_n v_{l,m} & \varphi_n v_{l',m'} & v_{l'',m''} & -v_{l'',m''} & v_{l''',m'''} & -v_{l''',m'''} \end{bmatrix}$ and the mapping from $i_{1,3}$ to (o_1, k_1) and (o_2, k_2) is given in Table 5.2.2.2.1a-3

$\upsilon = 8$	$W^{(8)}_{i_{1,1},o_1 i_{1,1,2}+k_1,o_1 i_{1,1,3}+k_1,o_1 i_{1,1,4}+k_1,i_{1,2},o_2 i_{1,2,2}+k_2,o_2 i_{1,2,3}+k_2,o_2 i_{1,2,4}+k_2,i_2}$
	where $W^{(8)}_{l,l',l'',l''',m,m',m'',m''',n} = \frac{1}{\sqrt{8P_{CSI-RS}}}\begin{bmatrix} v_{l,m} & v_{l,m} & v_{l',m'} & v_{l',m'} & v_{l'',m''} & v_{l'',m''} & v_{l''',m'''} & v_{l''',m'''} \\ \varphi_n v_{l,m} & -\varphi_n v_{l,m} & \varphi_n v_{l',m'} & -\varphi_n v_{l',m'} & v_{l'',m''} & -v_{l'',m''} & v_{l''',m'''} & -v_{l''',m'''} \end{bmatrix}$ and the mapping from $i_{1,3}$ to (o_1, k_1) and (o_2, k_2) is given in Table 5.2.2.2.1a-3

For a UE configured with higher layer parameter *typeI-codebookMode-r19* set to 'modeB', each PMI value corresponds to the codebook indices i_1, i_2, where the composite codebook index i_1 is defined by

$$i_1 = \begin{cases} [i_{1,1} \quad i_{1,2,1}] & \upsilon = 1 \\ [i_{1,1} \quad i_{1,2,1} \quad i_{1,2,2}] & \upsilon = 2 \\ [i_{1,1} \quad i_{1,2,1} \quad i_{1,2,2} \quad i_{1,2,3}] & \upsilon = 3 \\ [i_{1,1} \quad i_{1,2,1} \quad i_{1,2,2} \quad i_{1,2,3} \quad i_{1,2,4}] & \upsilon = 4 \\ [i_{1,1} \quad i_{1,2}] & \upsilon = 5,6,7,8 \end{cases}$$

and the composite codebook index i_2 is defined by

$$i_2 = \begin{cases} i_{2,1} & \upsilon = 1 \\ [i_{2,1} \quad i_{2,2}] & \upsilon = 2 \\ [i_{2,1} \quad i_{2,2} \quad i_{2,3}] & \upsilon = 3 \\ [i_{2,1} \quad i_{2,2} \quad i_{2,3} \quad i_{2,4}] & \upsilon = 4 \\ [i_{2,1} \quad i_{2,2} \quad i_{2,3}] & \upsilon = 5,6 \\ [i_{2,1} \quad i_{2,2} \quad i_{2,3} \quad i_{2,4}] & \upsilon = 7,8 \end{cases}$$

The codebook for 1-8 layers is given in Table 5.2.2.2.1a-7. The index $i_{1,1}$ is given by

$$i_{1,1} = [q_1 \quad q_2]$$

$$q_1 \in \{0,1,\ldots,O_1 - 1\}$$

$$q_2 \in \{0,1,\ldots,O_2 - 1\}$$

The index $i_{1,2,l}$, for $l = 1,\ldots,\upsilon$, is given by

$$i_{1,2,l} \in \{0,1,\ldots,N_1 N_2 - 1\}$$

and the mapping of $i_{1,1}$ and $i_{1,2,l}$ to $m_1^{(l)}$ and $m_2^{(l)}$ is given by

$$m_1^{(l)} = O_1 n_1^{(l)} + q_1$$

$$m_2^{(l)} = O_2 n_2^{(l)} + q_2$$

where

$$n_1^{(l)} = n^{(l)} \bmod N_1$$

$$n_2^{(l)} = \frac{n^{(l)} - n_1^{(l)}}{N_1}$$

$$n^{(l)} = N_1 N_2 - 1 - i_{1,2,l}$$

The index $i_{1,2}$ is given by

$$i_{1,2} \in \left\{0,1,\ldots,\binom{N_1 N_2}{L_G} - 1\right\}$$

where $L_G = 3$ for $\upsilon = 5,6$ and $L_G = 4$ for $\upsilon = 7,8$. The mapping of $i_{1,1}$ and $i_{1,2}$ to $m_1^{(g)}$ and $m_2^{(g)}$ for $g = 1,\ldots,L_G$ is obtained as in Clause 5.2.2.2.3, where the values of $C(x,y)$ are given in Table 5.2.2.2.5-4 and Table 5.2.2.2.1a-5.

The index $i_{2,l}$, for $l = 1, \ldots, \upsilon$ and $\upsilon = 1,2,3,4$ is given by

$$i_{2,l} \in \{0,1,2,3\}$$

and is mapped to $c_l = i_{2,l}$. The mapping of index $i_{2,g}$, for $g = 1, \ldots, L_G$ and $\upsilon = 5,6,7,8$, to c_l, with $l = 1, \ldots, \upsilon$, is given in Table 5.2.2.2.1a-6. The quantities φ_{c_l} and $v_{m_1^{(l)},m_2^{(l)}}$ for *typeI-codebookMode-r19* = 'modeB' are the same as defined above for 'modeA'.

Table 5.2.2.2.1a-5: Combinatorial coefficients $C(x, y)$

x \ y	1	2	3	4
19	19	171	969	3876
20	20	190	1140	4845
21	21	210	1330	5985
22	22	231	1540	7315
23	23	253	1771	8855
24	24	276	2024	10626
25	25	300	2300	12650
26	26	325	2600	14950
27	27	351	2925	17550
28	28	378	3276	20475
29	29	406	3654	23751
30	30	435	4060	27405
31	31	465	4495	31465
32	32	496	4960	35960
33	33	528	5456	40920
34	34	561	5984	46376
35	35	595	6545	52360
36	36	630	7140	58905
37	37	666	7770	66045
38	38	703	8436	73815
39	39	741	9139	82251
40	40	780	9880	91390
41	41	820	10660	101270

x \ y	1	2	3	4
42	42	861	11480	111930
43	43	903	12341	123410
44	44	946	13244	135751
45	45	990	14190	148995
46	46	1035	15180	163185
47	47	1081	16215	178365
48	48	1128	17296	194580
49	49	1176	18424	211876
50	50	1225	19600	230300
51	51	1275	20825	249900
52	52	1326	22100	270725
53	53	1378	23426	292825
54	54	1431	24804	316251
55	55	1485	26235	341055
56	56	1540	27720	367290
57	57	1596	29260	395010
58	58	1653	30856	424270
59	59	1711	32509	455126
60	60	1770	34220	487635
61	61	1830	35990	521855
62	62	1891	37820	557845
63	63	1953	39711	595665

Table 5.2.2.2.1a-6: Mapping of $i_{2,1}$, $i_{2,2}$, $i_{2,3}$, $i_{2,4}$ to $\{c_l\}$ for 5-8 layers and typeI-codebookMode-r19 = 'modeB'

$\upsilon = 5$		
$i_{2,1}$	c_1	c_2
0	0	2
1	1	3
$i_{2,2}$	c_3	c_4
0	0	2
1	1	3
$i_{2,3}$	c_5	
0	0	
1	1	

$\upsilon = 6$		
$i_{2,1}$	c_1	c_2
0	0	2
1	1	3
$i_{2,2}$	c_3	c_4
0	0	2
1	1	3
$i_{2,3}$	c_5	c_6
0	0	2
1	1	3

$\upsilon = 7$		
$i_{2,1}$	c_1	c_2
0	0	2
1	1	3
$i_{2,2}$	c_3	
0	0	
1	1	
2	2	
3	3	
$i_{2,3}$	c_4	c_5

$\upsilon = 8$		
$i_{2,1}$	c_1	c_2
0	0	2
1	1	3
$i_{2,2}$	c_3	c_4
0	0	2
1	1	3
$i_{2,3}$	c_5	c_6
0	0	2
1	1	3

2	2	
3	3	

0	0	2
1	1	3
$i_{2,4}$	c_6	c_7
0	0	2
1	1	3

$i_{2,4}$	c_7	c_8
0	0	2
1	1	3

Table 5.2.2.2.1a-7: Codebook for 1-8 layers and *typeI-codebookMode-r19* = 'modeB'

Layers	*typeI-codebookMode-r19* = 'modeB'
$\upsilon = 1$	$W^{(1)}_{m_1^{(1)},m_2^{(1)},c_1} = \frac{1}{\sqrt{P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} \end{bmatrix}$
$\upsilon = 2$	$W^{(2)}_{m_1^{(1)},m_1^{(2)},m_2^{(1)},m_2^{(2)},c_1,c_2} = \frac{1}{\sqrt{2P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(2)},m_2^{(2)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_2} v_{m_1^{(2)},m_2^{(2)}} \end{bmatrix}$
$\upsilon = 3$	$W^{(3)}_{m_1^{(1)},m_1^{(2)},m_1^{(3)},m_2^{(1)},m_2^{(2)},m_2^{(3)},c_1,c_2,c_3} = \frac{1}{\sqrt{3P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(3)},m_2^{(3)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_2} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_3} v_{m_1^{(3)},m_2^{(3)}} \end{bmatrix}$
$\upsilon = 4$	$W^{(4)}_{m_1^{(1)},m_1^{(2)},m_1^{(3)},m_1^{(4)},m_2^{(1)},m_2^{(2)},m_2^{(3)},m_2^{(4)},c_1,c_2,c_3,c_4} = \frac{1}{\sqrt{4P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(3)},m_2^{(3)}} & v_{m_1^{(4)},m_2^{(4)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_2} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_3} v_{m_1^{(3)},m_2^{(3)}} & \varphi_{c_4} v_{m_1^{(4)},m_2^{(4)}} \end{bmatrix}$
$\upsilon = 5$	$W^{(5)}_{m_1^{(1)},m_1^{(2)},m_1^{(3)},m_2^{(1)},m_2^{(2)},m_2^{(3)},c_1,c_2,c_3,c_4,c_5} = \frac{1}{\sqrt{5P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(3)},m_2^{(3)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_2} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_3} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_4} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_5} v_{m_1^{(3)},m_2^{(3)}} \end{bmatrix}$
$\upsilon = 6$	$W^{(6)}_{m_1^{(1)},m_1^{(2)},m_1^{(3)},m_2^{(1)},m_2^{(2)},m_2^{(3)},c_1,c_2,c_3,c_4,c_5,c_6} = \frac{1}{\sqrt{6P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(3)},m_2^{(3)}} & v_{m_1^{(3)},m_2^{(3)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_2} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_3} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_4} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_5} v_{m_1^{(3)},m_2^{(3)}} & \varphi_{c_6} v_{m_1^{(3)},m_2^{(3)}} \end{bmatrix}$
$\upsilon = 7$	$W^{(7)}_{m_1^{(1)},m_1^{(2)},m_1^{(3)},m_1^{(4)},m_2^{(1)},m_2^{(2)},m_2^{(3)},m_2^{(4)},c_1,c_2,c_3,c_4,c_5,c_6,c_7} = \frac{1}{\sqrt{7P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(3)},m_2^{(3)}} & v_{m_1^{(3)},m_2^{(3)}} & v_{m_1^{(4)},m_2^{(4)}} & v_{m_1^{(4)},m_2^{(4)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_2} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_3} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_4} v_{m_1^{(3)},m_2^{(3)}} & \varphi_{c_5} v_{m_1^{(3)},m_2^{(3)}} & \varphi_{c_6} v_{m_1^{(4)},m_2^{(4)}} & \varphi_{c_7} v_{m_1^{(4)},m_2^{(4)}} \end{bmatrix}$
$\upsilon = 8$	$W^{(8)}_{m_1^{(1)},m_1^{(2)},m_1^{(3)},m_1^{(4)},m_2^{(1)},m_2^{(2)},m_2^{(3)},m_2^{(4)},c_1,c_2,c_3,c_4,c_5,c_6,c_7,c_8} = \frac{1}{\sqrt{8P_{CSI-RS}}}\begin{bmatrix} v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(1)},m_2^{(1)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(2)},m_2^{(2)}} & v_{m_1^{(3)},m_2^{(3)}} & v_{m_1^{(3)},m_2^{(3)}} & v_{m_1^{(4)},m_2^{(4)}} & v_{m_1^{(4)},m_2^{(4)}} \\ \varphi_{c_1} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_2} v_{m_1^{(1)},m_2^{(1)}} & \varphi_{c_3} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_4} v_{m_1^{(2)},m_2^{(2)}} & \varphi_{c_5} v_{m_1^{(3)},m_2^{(3)}} & \varphi_{c_6} v_{m_1^{(3)},m_2^{(3)}} & \varphi_{c_7} v_{m_1^{(4)},m_2^{(4)}} & \varphi_{c_8} v_{m_1^{(4)},m_2^{(4)}} \end{bmatrix}$

For a UE configured with higher layer parameter *codebookType* set to 'typeI-SinglePanel-r19', the bitmap parameter *typeI-SinglePanel-ri-restriction-r19* forms the bit sequence $r_7, \ldots, r_1, r_0$, where r_0 is the LSB and r_7 is the MSB. When r_i is zero, $i \in \{0,1,\ldots,7\}$, PMI and RI reporting are not allowed to correspond to any precoder associated with $\upsilon = i + 1$ layers. The bitmap parameter *typeI-CBSR-r19* forms the bit sequence $a_{A_c-1}, \ldots, a_1, a_0$, where a_0 is the LSB and a_{A_c-1} is the MSB and the number of bits is given by $A_c = N_1 O_1 N_2 O_2 / \left(X_1^{(c)} X_2^{(c)}\right)$, where the values $\left(X_1^{(c)}, X_2^{(c)}\right) \in \{(1,1), (2,1), (2,2), (4,1), (4,2), (4,4)\}$ are configured by the higher layer parameters *valueOfX1-typeI-CBSR-r19* and

valueOfX2-typeI-CBSR-r19, respectively. A bit value of zero for bit $a_{\frac{N_2O_2}{X_2^{(c)}}x_1+x_2}$ indicates that PMI reporting is not allowed to correspond to any precoder associated with vector group $G_a(x_1,x_2)$, $x_1 = 0,\ldots,\frac{N_1O_1}{X_1^{(c)}}-1$, $x_2 = 0,\ldots,\frac{N_2O_2}{X_2^{(c)}}-1$, given by

$$G_a(x_1,x_2) = \left\{v_{X_1^{(c)}O_1x_1+l,X_2^{(c)}O_2x_2+m} : l = 0,1,\ldots,X_1^{(c)}O_1-1; m = 0,1,\ldots,X_2^{(c)}O_2-1\right\}.$$

The bitmap parameter *typeI-softScalingRanks1-2-r19* forms the bit sequence $b_{3A_s-1},\ldots,b_1,b_0$, where b_0 is the LSB and b_{3A_s-1} is the MSB and the number of bits is given by $B_s = 3N_1O_1N_2O_2/\left(X_1^{(s)}X_2^{(s)}\right)$, where the values $\left(X_1^{(s)},X_2^{(s)}\right) \in \{(2,1),(2,2),(4,1),(4,2),(4,4),(8,1)\}$ are configured by the higher layer parameters *valueOfX1-typeI-powerScaling-r19* and *valueOfX2-typeI-powerScaling-r19*, respectively. Bits $b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)+2}\,b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)+1}\,b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)}$ indicate a scaling factor, s_{x_1,x_2}, for vector group $G_s(x_1,x_2)$, $x_1 = 0,\ldots,\frac{N_1O_1}{X_1^{(s)}}-1$, $x_2 = 0,\ldots,\frac{N_2O_2}{X_2^{(s)}}-1$, given by

$$G_s(x_1,x_2) = \left\{v_{X_1^{(s)}x_1+l_s,X_2^{(s)}x_2+m_s} : l_s = 0,1,\ldots,X_1^{(s)}-1; m_s = 0,1,\ldots,X_2^{(s)}-1\right\}$$

The mapping from bits $b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)+2}\,b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)+1}\,b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)}$ to s_{x_1,x_2} is given in Table 5.2.2.2.1a-8.

Table 5.2.2.2.1a-8: Configurable scaling factor s_{x_1,x_2} for restricted vector group $G_s(x_1,x_2)$

Bits $b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)+2}\,b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)+1}\,b_{3\left(\frac{N_2O_2}{X_2^{(s)}}x_1+x_2\right)}$	Scaling factor s_{x_1,x_2}
000	$\frac{1}{4}$
001	$\frac{1}{2\sqrt{3}}$
010	$\frac{1}{2\sqrt{2}}$
011	$\frac{1}{\sqrt{6}}$
100	$\frac{1}{2}$
101	$\frac{1}{\sqrt{3}}$
110	$\frac{1}{\sqrt{2}}$
111	1

5.2.2.2.2 Type I Multi-Panel Codebook

For 8 antenna ports {3000, 3001, …, 3007}, 16 antenna ports {3000, 3001, …, 3015}, and 32 antenna ports {3000, 3001, …, 3031}, and the UE configured with higher layer parameter *codebookType* set to 'typeI-MultiPanel',

- The values of N_g, N_1 and N_2 are configured with the higher layer parameters *ng-n1-n2*. The supported configurations of $\left(N_g,N_1,N_2\right)$ for a given number of CSI-RS ports and the corresponding values of (O_1,O_2) are given in Table 5.2.2.2.2-1. The number of CSI-RS ports, $P_{\text{CSI-RS}}$, is $2N_gN_1N_2$.

- When $N_g = 2$, *codebookMode* shall be set to either '1' or '2'. When $N_g = 4$, *codebookMode* shall be set to '1'.

The bitmap parameter *ng-n1-n2* forms the bit sequence $a_{A_c-1},...,a_1,a_0$ where a_0 is the LSB and a_{A_c-1} is the MSB and where a bit value of zero indicates that PMI reporting is not allowed to correspond to any precoder associated with the bit. The number of bits is given by $A_c = N_1O_1N_2O_2$. Bit $a_{N_2O_2l+m}$ is associated with all precoders based on the quantity $v_{l,m}$, $l = 0,\ldots,N_1O_1-1$, $m = 0,\ldots,N_2O_2-1$, as defined below. The bitmap parameter *ri-Restriction* forms the bit sequence $r_3,...,r_1,r_0$ where r_0 is the LSB and r_3 is the MSB. When r_i is zero, $i \in \{0,1,\ldots,3\}$, PMI and RI reporting are not allowed to correspond to any precoder associated with $\upsilon = i+1$ layers.

Table 5.2.2.2.2-1: Supported configurations of (N_g, N_1, N_2) and (O_1, O_2)

Number of CSI-RS antenna ports, $P_{\text{CSI-RS}}$	(N_g, N_1, N_2)	(O_1, O_2)
8	(2,2,1)	(4,1)
16	(2,4,1)	(4,1)
	(4,2,1)	(4,1)
	(2,2,2)	(4,4)
32	(2,8,1)	(4,1)
	(4,4,1)	(4,1)
	(2,4,2)	(4,4)
	(4,2,2)	(4,4)

Each PMI value corresponds to the codebook indices i_1 and i_2, where i_1 is the vector

$$i_1 = \begin{cases} \begin{bmatrix} i_{1,1} & i_{1,2} & i_{1,4} \end{bmatrix} & \upsilon = 1 \\ \begin{bmatrix} i_{1,1} & i_{1,2} & i_{1,3} & i_{1,4} \end{bmatrix} & \upsilon \in \{2,3,4\} \end{cases}$$

and υ is the associated RI value. When *codebookMode* is set to '1', $i_{1,4}$ is

$$i_{1,4} = \begin{cases} i_{1,4,1} & N_g = 2 \\ \begin{bmatrix} i_{1,4,1} & i_{1,4,2} & i_{1,4,3} \end{bmatrix} & N_g = 4 \end{cases}.$$

When *codebookMode* is set to '2', $i_{1,4}$ and i_2 are

$$i_{1,4} = \begin{bmatrix} i_{1,4,1} & i_{1,4,2} \end{bmatrix}$$
$$i_2 = \begin{bmatrix} i_{2,0} & i_{2,1} & i_{2,2} \end{bmatrix}.$$

The mapping from $i_{1,3}$ to k_1 and k_2 for 2-layer reporting is given in Table 5.2.2.2.1-3. The mapping from $i_{1,3}$ to k_1 and k_2 for 3-layer and 4-layer reporting is given in Table 5.2.2.2.2-2.

- UE shall only use $i_{1,2} = 0$ and shall not report $i_{1,2}$ if the value of N_2 is 1.

Table 5.2.2.2.2-2: Mapping of $i_{1,3}$ to k_1 and k_2 for 3-layer and 4-layer CSI reporting

$i_{1,3}$	$N_1=2, N_2=1$		$N_1=4, N_2=1$		$N_1=8, N_2=1$		$N_1=2, N_2=2$		$N_1=4, N_2=2$	
	k_1	k_2	k_1	k_2	k_1	k_2	k_1	k_2	k_1	k_2
0	O_1	0	O_1	0	O_1	0	O_1	0	O_1	0
1			$2O_1$	0	$2O_1$	0	0	O_2	0	O_2
2			$3O_1$	0	$3O_1$	0	O_1	O_2	O_1	O_2
3					$4O_1$	0			$2O_1$	0

Several quantities are used to define the codebook elements. The quantities φ_n, a_p, b_n, u_m, and $v_{l,m}$ are given by

$$\varphi_n = e^{j\pi n/2}$$
$$a_p = e^{j\pi/4} e^{j\pi p/2}$$
$$b_n = e^{-j\pi/4} e^{j\pi n/2}$$
$$u_m = \begin{cases} \begin{bmatrix} 1 & e^{j\frac{2\pi m}{O_2 N_2}} & \cdots & e^{j\frac{2\pi m(N_2-1)}{O_2 N_2}} \end{bmatrix} & N_2 > 1 \\ 1 & N_2 = 1 \end{cases}$$
$$v_{l,m} = \begin{bmatrix} u_m & e^{j\frac{2\pi l}{O_1 N_1}} u_m & \cdots & e^{j\frac{2\pi l(N_1-1)}{O_1 N_1}} u_m \end{bmatrix}^T$$

Furthermore, the quantities $W_{l,m,p,n}^{1,N_g,1}$ and $W_{l,m,p,n}^{2,N_g,1}$ ($N_g \in \{2,4\}$) are given by

$$W_{l,m,p,n}^{1,2,1} = \frac{1}{\sqrt{P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} \\ \varphi_n v_{l,m} \\ \varphi_{p_1} v_{l,m} \\ \varphi_n \varphi_{p_1} v_{l,m} \end{bmatrix} \quad W_{l,m,p,n}^{2,2,1} = \frac{1}{\sqrt{P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} \\ -\varphi_n v_{l,m} \\ \varphi_{p_1} v_{l,m} \\ -\varphi_n \varphi_{p_1} v_{l,m} \end{bmatrix}$$

$$W_{l,m,p,n}^{1,4,1} = \frac{1}{\sqrt{P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} \\ \varphi_n v_{l,m} \\ \varphi_{p_1} v_{l,m} \\ \varphi_n \varphi_{p_1} v_{l,m} \\ \varphi_{p_2} v_{l,m} \\ \varphi_n \varphi_{p_2} v_{l,m} \\ \varphi_{p_3} v_{l,m} \\ \varphi_n \varphi_{p_3} v_{l,m} \end{bmatrix} \quad W_{l,m,p,n}^{2,4,1} = \frac{1}{\sqrt{P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} \\ -\varphi_n v_{l,m} \\ \varphi_{p_1} v_{l,m} \\ -\varphi_n \varphi_{p_1} v_{l,m} \\ \varphi_{p_2} v_{l,m} \\ -\varphi_n \varphi_{p_2} v_{l,m} \\ \varphi_{p_3} v_{l,m} \\ -\varphi_n \varphi_{p_3} v_{l,m} \end{bmatrix}$$

where

$$p = \begin{cases} p_1 & N_g = 2 \\ \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix} & N_g = 4 \end{cases},$$

and the quantities $W_{l,m,p,n}^{1,N_g,2}$ and $W_{l,m,p,n}^{2,N_g,2}$ ($N_g = 2$) are given by

$$W_{l,m,p,n}^{1,2,2} = \frac{1}{\sqrt{P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} \\ \varphi_{n_0} v_{l,m} \\ a_{p_1} b_{n_1} v_{l,m} \\ a_{p_2} b_{n_2} v_{l,m} \end{bmatrix} \quad W_{l,m,p,n}^{2,2,2} = \frac{1}{\sqrt{P_{\text{CSI-RS}}}} \begin{bmatrix} v_{l,m} \\ -\varphi_{n_0} v_{l,m} \\ a_{p_1} b_{n_1} v_{l,m} \\ -a_{p_2} b_{n_2} v_{l,m} \end{bmatrix}$$

where

$$p = \begin{bmatrix} p_1 & p_2 \end{bmatrix}$$
$$n = \begin{bmatrix} n_0 & n_1 & n_2 \end{bmatrix}.$$

The codebooks for 1-4 layers are given respectively in Tables 5.2.2.2.2-3, 5.2.2.2.2-4, 5.2.2.2.2-5, and 5.2.2.2.2-6.

Table 5.2.2.2.2-3: Codebook for 1-layer CSI reporting using antenna ports 3000 to 2999+$P_{CSI\text{-}RS}$

***codebookMode* = 1,** $N_g \in \{2,4\}$				
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,\ldots,N_g - 1$	i_2	
$0,\ldots,N_1O_1 - 1$	$0,\ldots,N_2O_2 - 1$	0,1,2,3	0,1,2,3	$W^{(1)}_{i_{1,1},i_{1,2},i_{1,4},i_2}$
where $W^{(1)}_{l,m,p,n} = W^{1,N_g,1}_{l,m,p,n}$.				

***codebookMode* = 2,** $N_g = 2$					
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,2$	$i_{2,0}$	$i_{2,q}$, $q = 1,2$	
$0,\ldots,N_1O_1 - 1$	$0,\ldots,N_2O_2 - 1$	0,1,2,3	0,1,2,3	0,1	$W^{(1)}_{i_{1,1},i_{1,2},i_{1,4},i_2}$
where $W^{(1)}_{l,m,p,n} = W^{1,N_g,2}_{l,m,p,n}$.					

Table 5.2.2.2.2-4: Codebook for 2-layer CSI reporting using antenna ports 3000 to 2999+$P_{CSI\text{-}RS}$

***codebookMode* = 1,** $N_g \in \{2,4\}$				
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,\ldots,N_g - 1$	i_2	
$0,\ldots,N_1O_1 - 1$	$0,\ldots,N_2O_2 - 1$	0,1,2,3	0,1	$W^{(2)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_{1,4},i_2}$
where $W^{(2)}_{l,l',m,m',p,n} = \frac{1}{\sqrt{2}}\begin{bmatrix} W^{1,N_g,1}_{l,m,p,n} & W^{2,N_g,1}_{l',m',p,n} \end{bmatrix}$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1-3.				

***codebookMode* = 2,** $N_g = 2$				
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,2$	$i_{2,q}$, $q = 0,1,2$	
$0,\ldots,N_1O_1 - 1$	$0,\ldots,N_2O_2 - 1$	0,1,2,3	0,1	$W^{(2)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_{1,4},i_2}$
where $W^{(2)}_{l,l',m,m',p,n} = \frac{1}{\sqrt{2}}\begin{bmatrix} W^{1,N_g,2}_{l,m,p,n} & W^{2,N_g,2}_{l',m',p,n} \end{bmatrix}$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.1-3.				

Table 5.2.2.2.2-5: Codebook for 3-layer CSI reporting using antenna ports 3000 to 2999+$P_{CSI\text{-}RS}$

***codebookMode* = 1,** $N_g \in \{2,4\}$				
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,\ldots,N_g - 1$	i_2	
$0,\ldots,N_1O_1 - 1$	$0,\ldots,N_2O_2 - 1$	0,1,2,3	0,1	$W^{(3)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_{1,4},i_2}$
where $W^{(3)}_{l,l',m,m',p,n} = \frac{1}{\sqrt{3}}\begin{bmatrix} W^{1,N_g,1}_{l,m,p,n} & W^{1,N_g,1}_{l',m',p,n} & W^{2,N_g,1}_{l,m,p,n} \end{bmatrix}$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.2-2.				

***codebookMode* = 2,** $N_g = 2$				
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,2$	$i_{2,q}$, $q = 0,1,2$	
$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1,2,3	0,1	$W^{(3)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_{1,4},i_2}$
where $W^{(3)}_{l,l',m,m',p,n} = \frac{1}{\sqrt{3}}\left[W^{1,N_g,2}_{l,m,p,n} \quad W^{1,N_g,2}_{l',m',p,n} \quad W^{2,N_g,2}_{l,m,p,n}\right]$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.2-2.				

Table 5.2.2.2.2-6: Codebook for 4-layer CSI reporting using antenna ports 3000 to 2999+$P_{CSI\text{-}RS}$

***codebookMode* = 1,** $N_g \in \{2,4\}$				
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,\ldots,N_g-1$	i_2	
$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1,2,3	0,1	$W^{(4)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_{1,4},i_2}$
where $W^{(4)}_{l,l',m,m',p,n} = \frac{1}{\sqrt{4}}\left[W^{1,N_g,1}_{l,m,p,n} \quad W^{1,N_g,1}_{l',m',p,n} \quad W^{2,N_g,1}_{l,m,p,n} \quad W^{2,N_g,1}_{l',m',p,n}\right]$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.2-2.				

***codebookMode* = 2,** $N_g = 2$				
$i_{1,1}$	$i_{1,2}$	$i_{1,4,q}$, $q = 1,2$	$i_{2,q}$, $q = 0,1,2$	
$0,\ldots,N_1O_1-1$	$0,\ldots,N_2O_2-1$	0,1,2,3	0,1	$W^{(4)}_{i_{1,1},i_{1,1}+k_1,i_{1,2},i_{1,2}+k_2,i_{1,4},i_2}$
where $W^{(4)}_{l,l',m,m',p,n} = \frac{1}{\sqrt{4}}\left[W^{1,N_g,2}_{l,m,p,n} \quad W^{1,N_g,2}_{l',m',p,n} \quad W^{2,N_g,2}_{l,m,p,n} \quad W^{2,N_g,2}_{l',m',p,n}\right]$ and the mapping from $i_{1,3}$ to k_1 and k_2 is given in Table 5.2.2.2.2-2.				

5.2.2.2.2a Refined Type I Multi-Panel Codebook

For 48, 64 and 128 antenna ports, obtained by aggregating K CSI-RS resources for channel measurement, and the UE configured with higher layer parameter *codebookType* set to 'typeI-MultiPanel-r19'

- The values of N_g, N_1 and N_2 are configured with the higher layer parameter *ng-n1-n2-r19*, where $N_g = K$ is the number of CSI-RS resources configured in the resource set for channel measurement. The supported configurations of (N_g, N_1, N_2), for a given number of CSI-RS ports, and the corresponding values of (O_1, O_2) are given in Table 5.2.2.2.2a-1. The number of CSI-RS ports, P_{CSI-RS}, is $2N_gN_1N_2$.

The bitmap parameter *ng-n1-n2-r19* forms the bit sequence $a_{A_c-1}, \ldots, a_1, a_0$, where a_0 is the LSB and a_{A_c-1} is the MSB and where a bit value of zero indicates that PMI reporting is not allowed to correspond to any precoder associated with the bit. The number of bits is given by $A_c = N_1O_1N_2O_2$. Bit $a_{N_2O_2l+m}$ is associated with all precoders based on the quantity $v_{l,m}$, $l = 0, \ldots, N_1O_1 - 1$, $m = 0, \ldots, N_2O_2 - 1$, as defined below. The bitmap parameter *typeI-MultiPanel-ri-restriction-r19* forms the bit sequence $r_3, \ldots, r_1, r_0$, where r_0 is the LSB and r_3 is the MSB. When r_i is zero, $i \in \{0,1, \ldots ,3\}$, PMI and RI reporting are not allowed to correspond to any precoder associated with $\upsilon = i + 1$ layers.

Table 5.2.2.2.2a-1: Supported configurations of (N_g, N_1, N_2) and (O_1, O_2)

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_g, N_1, N_2)	(O_1, O_2)
48	(2,4,3)	(4,4)
	(2,6,2)	(4,4)
64	(2,4,4)	(4,4)